ORFfinder Home - NCBI

https://www.ncbi.nlm.nih.gov/orffinder/

Chat

Open Reading Frame Finder



ORF finder searches for open reading frames (ORFs) in the DNA sequence you enter. The program returns the range of each ORF, along with its protein translation. Use ORF finder to search newly sequenced DNA for potential protein encoding segments, verify predicted protein using SMART BLAST or BLASTP.

This web version of the ORF finder is limited to the subrange of the query sequence up to 50 kb long. Stand-alone version, which doesn't have query sequence length limitation, is available for Linux x64.

Examples (click to set values, then click Submit button) :

- NC_011604 Salmonella enterica plasmid pWES-1; genetic code: 11; 'ATG' and alternative initiation codons; minimal ORF length: 300 nt
- NM_000059; genetic code: 1; start codon: 'ATG only'; minimal ORF length: 150 nt

Enter Query Sequence

Enter accession number, gi, or nucleotide sequence in FASTA format:

```
TGTATGTTTGGAGAAGGTCCCTCTGGAGGCCTTCCTGGCAGACAGGGGATTGGATTCAGG
CTGTGGAAGCAGGACGGTAGGGGGTGTGATTCCAGGATGTGGAAAGGAATAAAAATGAA
GAGCCCCGGGGAAGAGGTCAAGGGAGTTGGGGGACCCGAGTTCCTGGCTCCAGGGGGAAG
CGAGTGGTAAGTCTGTGAACAGAGCCCAGCTGTGGATTCTGTCAATGGGGTCAGGTCTCA
CCCTGTGGCTTCCAGGGCAGCAAGGCAGGAAGGAGGCGTCTGCCACAAGGCCAGCTTCCT
GGGGCCAGAGCCGTGAAGGCCCAGGGGACCTGCGTGTCTTGGCTCCACGCCAGATGTGTT
ATTATTTATGTCTCTGAGAATGTCTGGATCTCAGAGCCGAATTACAATAAAAACATCTTT
AAACTT
```

From: To:

Choose Search Parameters

Minimal ORF length (nt): 75

Genetic code: 1. Standard

ORF start codon to use:

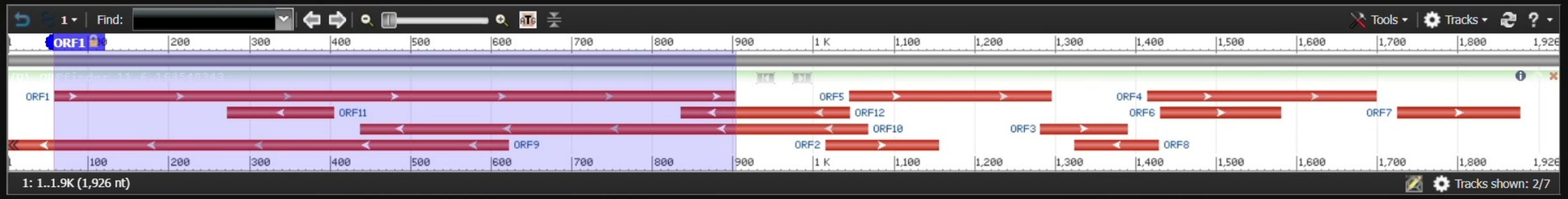
"ATG" only

ORFfinder Viewer - NCBI

https://www.ncbi.nlm.nih.gov/orffinder/

Chat

ORFs found: 12 Genetic code: 1 Start codon: 'ATG' only



Six-frame translation...

ORF1 (281 aa) Display ORF as... **Mark**

```
>lcl|ORF1
MPAVKKEFPGREDLALALATFHPTLAALPLPPLPGYLAPLPAAAALPPAA
SLPASAAGYEALLAPPLRPPRAYLSLHEAAPHLHLPRDPLALERFSATAA
AAPDFQPLLDNGEPCIEVECGANRALLYVRKLCQGSKGPSIRHRGEWLTP
NEFQFVSGRETAKDWKRSIRHKGKSLKTLMSKGILQVHPPICDCPGCRIS
SPVNRGRLADKRTVALPAARNLKKERTPSFSASDGDSDGSGPTCGRRPGL
KQEDGPHIRIMKRRVHTHWDVNISFREASCR
```

ORF1

SmartBLAST

BLAST

Marked set (0)

SmartBLAST best hit titles...

BLAST

BLAST Database:

UniProtKB/Swiss-Prot (swissprot)

Mark subset... Marked: 0 Download marked set as Protein FASTA

Label	Strand	Frame	Start	Stop	Length (nt | aa)
ORF1	+	1	58	903	846 | 281
ORF10		2	1067	438	630 | 209
ORF9	-	1	621	>1	621 | 206
ORF4	+	1	1414	1698	285 | 94
ORF5	+	3	1044	1295	252 | 83
ORF12	-	3	1045	836	210 | 69
ORF7	+	3	1725	1877	153 | 50
ORF6	+	3	1431	1580	150 | 49
ORF2	+	1	1015	1155	141 | 46
ORF11	-	2	404	273	132 | 43
ORF3	+	1	1282	1389	108 | 35

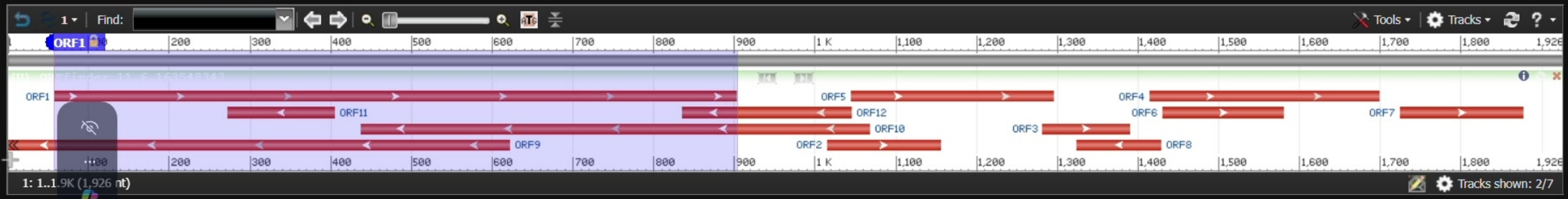
Go back to the submitting page...

ORFfinder Viewer - NCBI

https://www.ncbi.nlm.nih.gov/orffinder/

Chat

ORFs found: 12 Genetic code: 1 Start codon: 'ATG' only



Six-frame translation...

ORF1 (281 aa) Display ORF as... Mark

>lcl|ORF1
MPAVKKEFPGREDLALALATFHPTLAALPLPPLPGYLAPLPAAAALPPAA
SLPASAAGYEALLAPPLRPPRAYLSLHEAAPHLHLPRDPLALERFSATAA
AAPDFQPLLDNGEPCIEVECGANRALLYVRKLCQGSKGPSIRHRGEWLTP
NEFQFVSGRETAKDWKRSIRHKGKSLKTLMSKGILQVHPPICDCPGCRIS
SPVNRGRLADKRTVALPAARNLKKERTPSFSASDGDSDGSGPTCGRRPGL
KQEDGPHIRIMKRRVHTHWDVNISFREASCR

ORF1

SmartBLAST

BLAST

Marked set (0)

SmartBLAST best hit titles...

BLAST

BLAST Database:

UniProtKB/Swiss-Prot (swissprot)

Mark subset... Marked: 0 Download marked set as Protein FASTA

Label	Strand	Frame	Start	Stop	Length (nt | aa)
ORF1	+	1	58	903	846 | 281
ORF10	-	2	1067	438	630 | 209
ORF9	-	1	621	>1	621 | 206
ORF4	+	1	1414	1698	285 | 94
ORF5	+	3	1044	1295	252 | 83
ORF12	-	3	1045	836	210 | 69
ORF7	+	3	1725	1877	153 | 50
ORF6	+	3	1431	1580	150 | 49
ORF2	+	1	1015	1155	141 | 46
ORF11	-	2	404	273	132 | 43
ORF3	+	1	1282	1389	108 | 35

Go back to the submitting page...

Translate tool

Translate is a tool which allows the translation of a nucleotide (DNA/RNA) sequence to a protein sequence.

DNA or RNA sequence

```
GGGACCCGAGTTCCTGGCTCCAGGGGGAAG
CGAGTGGTAAGTCTGTGAACAGAGCCCAGCT
GTGGATTCTGTCAATGGGGTCAGGTCTCA
CCCTGTGGCTTCCAGGGCAGCAAGGCAGGAA
GGAGGCGTCTGCCACAAGGCCAGCTTCCT
GGGGCCAGAGCCGTGAAGGCCCAGGGGACC
TGCGTGTCTTGGCTCCACGCCAGATGTGTT
ATTATTTATGTCTCTGAGAATGTCTGGATCTCAG
AGCCGAATTACAATAAAAACATCTTT
AAACTT
```

Output format

- ○ Verbose: Met, Stop, spaces between residues
- ◉ Compact: M, -, no spaces
- ○ Includes nucleotide sequence
- ○ Includes nucleotide sequence, no spaces

DNA strands

☑ forward ☑ reverse

Genetic codes - See NCBI's genetic codes

Standard, only A,C,G,T

reset | TRANSLATE!

Standard, only A,C,G,T

reset | TRANSLATE!

Results of translation

- Open reading frames are highlighted in red
- Select your initiator on one of the following frames to retrieve your amino acid sequence

Download all the translated frames

5'3' Frame 1

AGISCCRRRRAGLRSHRGAMPAVKKEFPGREDLALALATFHPTLAALPLPPLPGYLAPLPAAAALPPAASLPASAAGYEALLAPPLRPP
RAYLSLHEAAPHLHLPRDPLALERFSATAAAAPDFQPLLDNGEPCIEVECGANRALLYVRKLCQGSKGPSIRHRGEWLTPNEFQFVSGR
ETAKDWKRSIRHKGKSLKTLMSKGILQVHPPICDCPGCRISSPVNRGRLADKRTVALPAARNLKKERTPSFSASDGDSDGSGPTCGRRP
GLKQEDGPHIRIMKRRVHTHWDVNISFREASCR-EPCCACIRGGRDSPPSLPPLTVPCCLLSAVSA-ADAVMLAECLPGLTCAARLSPI
PGGRQCCTQLGLRDPRPSQMQLSAAAQV-VLGSPALELECTSSHVRLAI-AQK-GCHLPLPTLLETHKEVRMLMCEWGPEGCVGALRRG
DVCRVEEQGRACRRDEP-TWQLWFSWMLKSHGCWACGHRPPGGSHRKFLSRAVTRCMFGEGPSGGLPGRQGIGFRLWKQDGRGCDSRMW
KGDKNEEPRGRGQGSWGTRVPGSRGKRVVSL-TEPSCGFCQWGQVSPCGFQGSKAGRRRLPQGQLPGARAVKAQGTCVSWLHARCVIIY
VSENVWISEPNYNKNIFKL

5'3' Frame 2

PE-VAAAGGERGCGATGAPCRRSRRSSRAARTWPWLWPRSTRPWPRCRCRRCQATWRHCPRRPPSPRPPRCPPRPPVTRLCWPRRSAPR
APTSACTRPPRTSTCPGTRWPSSASRPPRPRPRISSRCWTTASRASRWSAAPTARCSTCANSARAARARPSATAASGSRPTSSSSSAAA
RRPRTGSAASATKGKV-RRLCPRGSCRCILRSATARAAEYPPR-TGGGWQTRGQSPCLPPGT-RRSELPASLPAMVTATGVAPPVGGGQ
A-SRRMVRTSVS-REESTPTGT-TSLSERRPAGRSRAVRA-EGAVTPLPPSHP-PCPAVCCPLSQRELMD-CWLSVCQV-HVLQGCPPS
READSVAPSWD-GTPDPVRCSSRQQLRCEFWAARPWS-SALPPM-DWPFEPKNEAVTSPFPPS-RPTRR-EC-CVSGALKGV-EL-GEG
MSAE-RNREGRVGGTRSEPGSSGSVGC-RVMDAGPVGTVLQAGATESS-AGQ-PGVCLEKVPLEAFLADRGLDSGCGSRTVGGVIPGCG
KEIKMKSPGEEVKGVGGPEFLAPGGSEW-VCEQSPAVDSVNGVRSHPVASRAARQEGGVCHKASFLGPEP-RPRGPACLGSTPDVLLFM

European Bioinformatics Institute (EMBL-EBI)

Scheduled System Maintenance
We have scheduled system maintenance from 8:00 AM GMT on Tuesday, 3 November 2026, until 12:00 PM GMT on Thursday, 5 November 2026. During this period, the Job Dispatcher service will operate at reduced capacity. You can continue submitting jobs as normal; however, jobs running or queued before the migration may be lost. Please plan your workloads accordingly. If you need assistance or have feedback, please contact us.

Sequence Translation

Sequence Translation is used to translate nucleic acid sequence to corresponding peptide sequences.
Back-translation is used to predict the possible nucleic acid sequence that a specified peptide sequence has originated from.

Nucleotide Sequence Translation

EMBOSS Transeq

EMBOSS Transeq translates nucleic acid sequences to the corresponding peptide sequences.

Launch EMBOSS Transeq

EMBOSS Sixpack

EMBOSS Sixpack displays DNA sequences with 6-frame translation and ORFs.

Launch EMBOSS Sixpack

Protein Sequence Back-

EMBOSS Backtranseq

three forward and three reverse frames, and output multiple frame translations at once.

Input sequence

Paste your sequence here - or use the example sequence

```
CTGTGGAAGCAGGACGGTAGGGGGTGTGATTCCAGGATGTGGAAAGGAGATAAAAATGAA
GAGCCCCGGGGAAGAGGTCAAGGGAGTTGGGGGACCCGAGTTCCTGGCTCCAGGGGGAAG
CGAGTGGTAAGTCTGTGAACAGAGCCCAGCTGTGGATTCTGTCAATGGGGTCAGGTCTCA
CCCTGTGGCTTCCAGGGCAGCAAGGCAGGAAGGAGGCGTCTGCCACAAGGCCAGCTTCCT
GGGGCCAGAGCCGTGAAGGCCCAGGGGACCTGCGTGTCTTGGCTCCACGCCAGATGTGTT
ATTATTTATGTCTCTGAGAATGTCTGGATCTCAGAGCCGAATTACAATAAAAACATCTTT
AAACTT
```

Choose File **No file chosen**

Use the example **Clear sequence** More example inputs

Parameters

FRAME

1

CODON TABLE

Standard Code

More options

Submit

Title

EMBOSS TRANSEQ's job

Submit

Parameters

CODON TABLE

Standard Code

FIRST ORF

yes

LAST ORF

yes

REVERSE

yes

ORF MINIMUM SIZE

1

Less options

Submit

Title

EMBOSS SIXPACK's job

Submit

Tool output

EMBOSS_001

Download

```
  A  G  I  S  C  C  R  R  R  R  A  G  L  R  S  H  R  G  A  M     F1
   P  E  *  V  A  A  A  G  G  E  R  G  C  G  A  T  G  A  P  C    F2
    R  N  K  L  L  P  P  A  E  S  G  A  A  E  P  P  G  R  H  A   F3
1 GCCGGAATAAGTTGCTGCCGCCGGCGGAGAGCGGGGCTGCGGAGCCACCGGGGCGCCATG 60
  ----:----|----:----|----:----|----:----|----:----|----:----|
1 CGGCCTTATTCAACGACGGCGGCCGCCTCTCGCCCCGACGCCTCGGTGGCCCCGCGGTAC 60
   A  P  I  L  Q  Q  R  R  R  L  A  P  S  R  L  W  R  P  A  M   F6
  X  R  F  L  N  S  G  G  A  S  L  P  A  A  S  G  G  P  R  W    F5
   G  S  Y  T  A  A  A  P  P  S  R  P  Q  P  A  V  P  A  G  H   F4


  P  A  V  K  K  E  F  P  G  R  E  D  L  A  L  A  L  A  T  F    F1
   R  R  S  R  R  S  S  R  A  A  R  T  W  P  W  L  W  P  R  S   F2
    G  G  Q  E  G  V  P  G  P  R  G  P  G  P  G  S  G  H  V  P  F3
61 CCGGCGGTCAAGAAGGAGTTCCCGGGCCGCGAGGACCTGGCCCTGGCTCTGGCCACGTTC 120
  ----:----|----:----|----:----|----:----|----:----|----:----|
61 GGCCGCCAGTTCTTCCTCAAGGGCCCGGCGCTCCTGGACCGGGACCGAGACCGGTGCAAG 120
   G  A  T  L  F  S  N  G  P  R  S  S  R  A  R  A  R  A  V  N   F6
  A  P  P  *  S  P  T  G  P  G  R  P  G  P  G  P  E  P  W  T    F5
   R  R  D  L  L  L  E  R  A  A  L  V  Q  G  Q  S  Q  G  R  E  F4
```

```
          *  V                                                F5
             F  K                                             F4
```

```
#################################
Minimum size of ORFs : 1

Total ORFs in frame 1 :     8
Total ORFs in frame 2 :    26
Total ORFs in frame 3 :    15
Total ORFs in frame 4 :     7
Total ORFs in frame 5 :    11
Total ORFs in frame 6 :    14

Total ORFs :    81
#################################
```

Coloured sequence(s)

Show

If you use this service, please consider citing the following publication: The EMBL-EBI Job Dispatcher sequence analysis tools framework in 2024. More information about this bioinformatics application can be found in its bio.tools record.

Please read the provided Help & Privacy before seeking help from our support staff. If you have any feedback or experienced any issues please let us know via EMBL-EBI Support. Read our Privacy Notice if you are concerned with your privacy and how we handle personal information.

```
     P C G F Q G S K A G R R R L P Q G Q L P   F1
      P V A S R A A R Q E G G V C H K A S F L  F2
       L W L P G Q Q G R K E A S A T R P A S W F3
1741CCCTGTGGCTTCCAGGGCAGCAAGGCAGGAAGGAGGCGTCTGCCACAAGGCCAGCTTCCT1800
    ----:----|----:----|----:----|----:----|----:----|----:----|
1741GGGACACCGAAGGTCCCGTCGTTCCGTCCTTCCTCCGCAGACGGTGTTCCGGTCGAAGGA1800
     G Q P K W P L L A P L L R R G C P W S G   F6
     V R H S G P C C P L F S A D A V L G A E   F5
      G T A E L A A L C S P P T Q W L A L K R F4


     G A R A V K A Q G T C V S W L H A R C V   F1
      G P E P * R P R G P A C L G S T P D V L  F2
       G Q S R E G P G D L R V L A P R Q M C Y F3
1801GGGGCCAGAGCCGTGAAGGCCCAGGGGACCTGCGTGTCTTGGCTCCACGCCAGATGTGTT1860
    ----:----|----:----|----:----|----:----|----:----|----:----|
1801CCCCGGTCTCGGCACTTCCGGGTCCCCTGGACGCACAGAACCGAGGTGCGGTCTACACAA1860
     P A L A T F A W P V Q T D Q S W A L H T   F6
     Q P W L R S P G P S R R T K A G R W I H   F5
      P G S G H L G L P G A H R P E V G S T N F4


     I I Y V S E N V W I S E P N Y N K N I F   F1
      L F M S L R M S G S Q S R I T I K T S L  F2
       Y L C L * E C L D L R A E L Q * K H L * F3
1861ATTATTTATGTCTCTGAGAATGTCTGGATCTCAGAGCCGAATTACAATAAAAACATCTTT1920
    ----:----|----:----|----:----|----:----|----:----|----:----|
1861TAATAAATACAGAGACTCTTACAGACCTAGAGTCTCGGCTTAATGTTATTTTTGTAGAAA1920
     I I * T E S F T Q I E S G F * L L F M K   F6
     * * K H R Q S H R S R L A S N C Y F C R   F5
      N N I D R L I D P D * L R I V I F V D K F4
```